SVD

- SVD of $\boxed{A_{m\times n} = U_{m\times m}\,\Sigma_{m\times n}\,V^T_{n\times n}}$

  orthogonal

- calculate AA^T then do $AA^T - \lambda I = 0$. find λs.

  eigenvalue

- from λs, find $\sigma_n = \sqrt{\lambda_n}$.
- for each eigenvalue find corresponding eigenvector.

$$\begin{pmatrix} \lambda_1 \to v_1 \\ \lambda_2 \to v_2 \\ \vdots \end{pmatrix} \cdot (AA^T - \lambda I) \times \begin{bmatrix} x \\ y \\ \vdots \end{bmatrix} = 0$$

from this operation, you get equations of x & y. write their simplest relation as the eigenvector.

eg: $7x - 28y = 0 \Rightarrow x = 4y \quad \therefore v = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$

then normalise this eigenvector to make it's length 1.

eg: $\begin{bmatrix} 1 \\ 4 \end{bmatrix} \to \sqrt{1^2 + 4^2} = \sqrt{17} \Rightarrow v = \frac{1}{\sqrt{17}}\begin{bmatrix} 1 \\ 4 \end{bmatrix}$

- then, write U with eigenvectors as the colums. Higher eigenvalue's eigenvectors come first.

eg: $U = \begin{bmatrix} v_1 & v_2 \end{bmatrix}$

$\lambda_1 > \lambda_2$

- Repeat the process with A^TA to form V^T.

  then, write V^T with eigenvectors as the <u>rows</u>. Higher eigenvalue's eigenvectors come first.

- To find Σ, square root of eigenvalues (obtained from U) ie, $\sigma_1, \sigma_2 \ldots$ are put as diagonal elements in descending order.

  Σ is a $m\times n$ matrix where all elements are 0 except the diagonal elements.

Solved Example

Q. find SVD of $A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}$.

Sol^n: $A^T = \begin{bmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{bmatrix}$

1. $AA^T = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{bmatrix} = \begin{bmatrix} 17 & 8 \\ 8 & 17 \end{bmatrix}$

$AA^T - \lambda I \Rightarrow \begin{bmatrix} 17-\lambda & 8 \\ 8 & 17-\lambda \end{bmatrix}$

$= (17-\lambda)^2 - 64 = 0$

$= \lambda^2 - 34\lambda + 289 - 64 = 0$

$\vdots$

$\lambda_1 = 25 \; ; \; \lambda_2 = 9.$

$\sigma_1 = 5; \; \sigma_2 = 3.$

for λ_1: $\begin{bmatrix} -8 & 8 \\ 8 & -8 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$-8x + 8y = 0 \quad \therefore x = y$

$\therefore v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \rightarrow \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

for λ_2: $\begin{bmatrix} 8 & 8 \\ 8 & 8 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$8x + 8y = 0 \quad \therefore x = -y$

$\therefore v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \rightarrow \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$$\boxed{\therefore U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}}$$

2. $A^TA = \begin{bmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix} = \begin{bmatrix} 13 & 12 & 2 \\ 12 & 13 & -6 \\ 2 & -2 & 8 \end{bmatrix}$

$\begin{bmatrix} 13-\lambda & 12 & 2 \\ 12 & 13-\lambda & -6 \\ 2 & -2 & 8-\lambda \end{bmatrix} = 0$

solve the det.

we get a cubic with

$\lambda_1 = 25, \; \lambda_2 = 9, \; \lambda_3 = 0.$

for λ_1: $\begin{bmatrix} -12 & 12 & 2 \\ 12 & -12 & -6 \\ 2 & -2 & -17 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$\Rightarrow -12x + 12y + 2z = 0$

$\therefore -6x + 6y + z = 0.$

$12x - 12y - 6z = 0$

$\therefore 2x - 2y - z = 0$

$2x - 2y - 17z = 0$

$-6x + 6y + z = 0$

$6x - 6y - 3z = 0$

$\therefore z = 0$

$\therefore x = y$

$\therefore V_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

$V_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

for λ_2: (9) $\begin{bmatrix} 4 & 12 & 2 \\ 12 & 4 & -6 \\ 2 & -2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$\Rightarrow 4x + 12y + 2z = 0$

$\therefore 2x + 6y + z = 0$

$12x + 4y - 6z = 0$

$6x + 2y - 3z = 0$

$2x - 2y - z = 0.$

$x = -y.$

$2x + 6y + z = 0$

$-2x + 2y + z = 0$

$8y + 2z = 0$

$4x = z$

$V_2 = \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix} \rightarrow \begin{bmatrix} \frac{1}{3\sqrt{2}} \\ -1/3\sqrt{2} \\ 4/3\sqrt{2} \end{bmatrix}$

for λ_3: (0) $\begin{bmatrix} 13 & 12 & 2 \\ 12 & 13 & -6 \\ 2 & -2 & 8 \end{bmatrix}$

$V_3 = \begin{bmatrix} 2/3 \\ -2/3 \\ -1/3 \end{bmatrix}$

$$\boxed{V^T = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/3\sqrt{2} & -1/3\sqrt{2} & 4/3\sqrt{2} \\ 2/3 & -2/3 & -1/3 \end{bmatrix}}$$

3. $$\boxed{\Sigma = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \end{bmatrix}}$$

$\therefore A = U \Sigma V^T.$: Ans